AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION

Test Report No. 2023 – 0008

Agricultural Machinery – Small Engine
KRESS KE215R

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Test Report Number	2023 – 0008
Date of Issuance	January 20, 2023
Valid Until	January 20, 2028
Agricultural Machinery	Small Engine
Type	Four-stroke cycle, Air-cooled, Single inclined cylinder, Gasoline engine
Brand	KRESS
Model	KE215R
Test Requested by	JGO VENTURES CORPORATION #6, 4th Flr., Don C. Manuel St., Kaingin Rd., Quezon City

SUMMARY

Performance Criteria	Requirement as per PAES 116:2001	AMTEC Test
Maximum Power as Percentage of Rated Maximum Power, %	80	81.08[a]
Continuous Power as Percentage of Rated Maximum Power, %	80	77.10[b]
Noise Level, dB(A), maximum	92[c]	89.4

[a] Occurred at 2003 rpm, 43.28 N-m torque, 9.07 kW power developed, and 4.73 L/h fuel consumption rate during varying load test; Manufacturer's maximum rated power was 11.19 kW at 1800 rpm.

[b] Average output power during the five-hour continuous running test was 8.63 kW at shaft speed of 1803 rpm and torque of 45.71 N-m; Manufacturer's continuous rated power was not provided.

[c] Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor. Philippines, 1983.

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Not provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	2022-0351
3. Manufacturer	POSITEC TECHNOLOGY CO. LTD. 18 Dongwang Rd., Shuzhou, Industrial Park, China
4. Starting the engine	Easy (see Starting Test result)
5. During varying speed test	No breakdown occurred.
6. During continuous running test	Engine oil leak and protruding gasket at the crank case were observed on the 4th hour of the continuous running test. (Figure 7)
7. Fuel tank capacity shall have a capacity which shall not require replenishment of fuel for at least two hours of operation at rated continuous power	Fuel tank capacity allows for continuous running operation of 1.45 hours before replenishing.
8. The base shall be durable and have provision for easy mounting of the engine	Provided
9. There shall be provisions that will orient the exhaust away from operator	Not provided
10. The engine shall be equipped with cooling system suitable for tropical operations	The engine is air-cooled.
11. Castings shall be free of shrink holes, blowholes, cracks, scales, blisters and other similar injurious defects. The surface of castings shall be cleaned by sandblasting, shot blasting, picking or any standard method	The engine is free from manufacturing defects.
12. The engine shall be free from sharp edges and surfaces that may injure the operator	No sharp edges and surfaces were observed.
13. Any uncoated metallic surfaces shall be free from rust and shall be painted properly	Uncoated surfaces are free from rust and are properly painted.

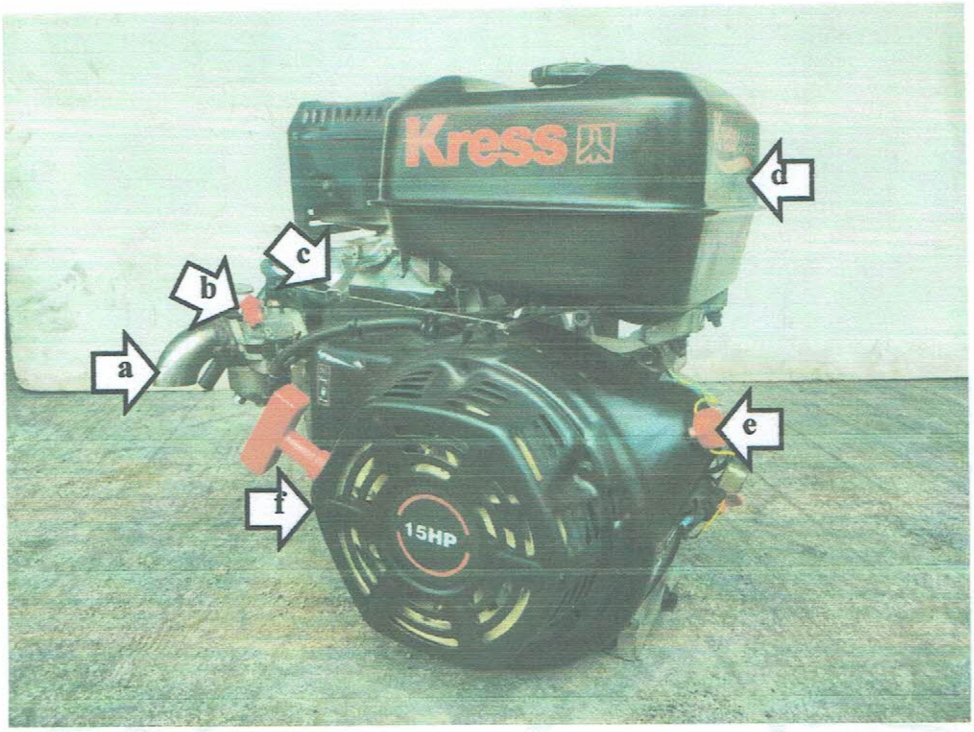
PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	January 16 and 17, 2023
Location of Test	AMTEC Test Laboratory, UPLB, College, Los Baños, Laguna
Items Tested	Operator's Manual, Starting Test, Varying Load Test, and Continuous Running Test

METHODS OF TEST

The small engine was tested based on the PAES 117:2000 Agricultural Machinery – Small Engine – Methods of Test.

DESCRIPTION OF THE ENGINE



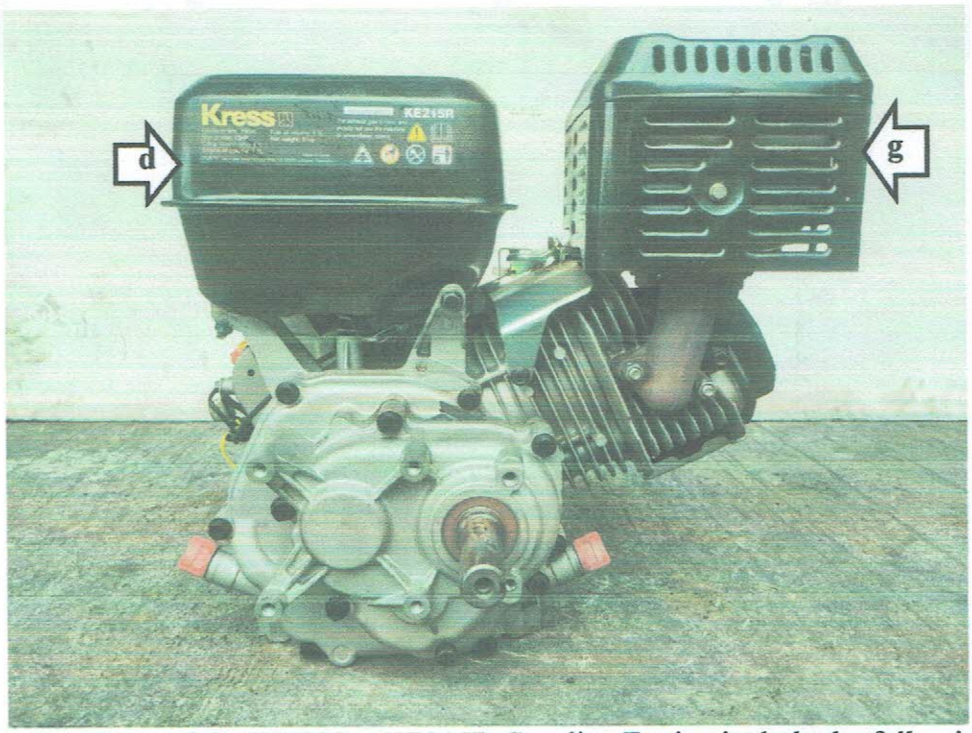


Figure 1. Parts of the KRESS – KE215R Gasoline Engine include the following: (a) Air cleaner stainless pipe, (b) Choke, (c) Throttle lever, (d) Fuel tank, (e) On/off switch, (f) Rope-recoil, and (g) Muffler.

DESCRIPTION OF THE ENGINE (Cont'n)

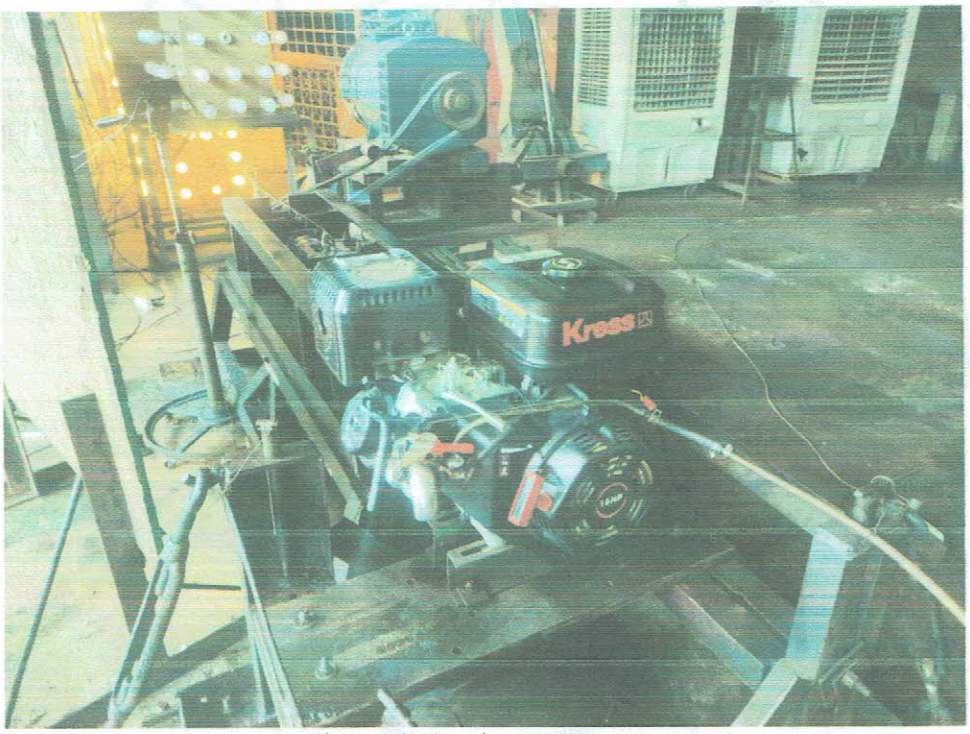


Figure 2. Engine test set-up.

SPECIFICATIONS

	Item	Manufacturer's Specification[a]	AMTEC Verification
1	Dimensions and weight of engine		
1.1	Overall length, mm	480	460
1.2	Overall width, mm	405	420
1.3	Overall height, mm	443	425
1.4	Weight of the engine, kg	31	33.0
2	Engine		
2.1	Brand	KRESS	KRESS
2.2	Model	KE215R	KE215R
2.3	Serial number	ND	20210201489272
2.4	Make	ND	ND
2.5	Type		
2.5.1	Based on number of strokes to complete one cycle	Four-stroke cycle	Four-stroke cycle
2.5.2	Based on number of cylinders	One	One
2.5.3	Based on cylinder arrangement	Inclined	Inclined
2.6	Maximum brake power at rpm, kW	11.19 kW at 1800 rpm	9.07 kW at 2003 rpm
2.7	Continuous brake power at rpm, kW	ND	8.63 kW at 1803 rpm
2.8	Bore × stroke, mm	64 × 88	NM
2.9	Displacement volume, cm^3	390	NM
2.1	Fuel system		
2.10.1	Fuel used	Gasoline	Gasoline
2.10.2	Type of fuel feed	ND	Gravity
2.10.3	Tank capacity, L	6.5	6.1
2.11	Cooling system	Air-cooled	Air-cooled
2.12	Lubrication system	ND	ND
2.13	Starting system	Recoil / electric start	Rope-recoil starter
2.14	Governor system	ND	ND
2.15	Air cleaner	Semi-dry	Cut stainless pipe
2.16	Exhaust system	Exhaust pipe	Muffler
2.17	Other attachments/accessories	None	None

[a] The data were obtained from the operator's manual provided by the requesting party.
[b] Weight includes engine oil only
Note: ND – No Data, NM – Not Measured

RESULTS

Starting Test

Table 1. Results of starting test of KRESS KE215R Gasoline Engine.

Item	Cold Starting	Hot Starting
Test condition		
Ambient Temperature, °C	26.2	28.0
Relative Humidity, %	90.4	83.2
Fuel temperature	NM	NM
Engine oil temperature	24.9	124.4
Number of attempts to start the engine	2	2
Ease of cranking/starting	Easy	Easy

Note: NM – Not Measured

RESULTS (Cont'n)

Varying Load Test

Table 2. Test conditions during the varying load test of KRESS KE215R Gasoline Engine.

Date of test	January 16, 2023
Ambient temperature, °C	28.0
Relative humidity, %	84.6

Table 3. Results of varying load performance test of the KRESS KE215R Gasoline Engine with maximum power developed at 81.08% of the rated maximum power.

Engine Shaft Speed, rpm	Shaft Torque, N-m	Output Power, kW	Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h	Noise Level, dB(A)	Temperature, °C	
						Engine Oil	Exhaust Air
2143	11.32	2.54	2.21	630.56	86.9	74.9	404.6
2102	21.07	4.64	3.08	480.99	88.3	85.6	447.9
2003	43.28	9.07	4.73	377.84	89.0	97.2	578.0
1905	45.16	9.00	4.46	359.52	88.4	105.3	560.6
1806	45.62	8.62	4.20	353.50	88.5	112.6	557.3
1706	46.56	8.31	4.00	349.23	87.5	117.1	540.2
1607	47.06	7.91	3.95	362.27	87.3	121.2	527.5
1506	46.90	7.39	3.63	355.95	86.5	124.1	494.9
1406	46.79	6.88	3.32	349.34	86.2	126.4	466.1
1308	46.54	6.37	3.16	359.32	85.7	127.5	447.5
1277	46.49	6.21	3.06	357.38	88.6	131.6	436.6

Note: Engine was set at full throttle. No breakdown occurred. Noise level measured at 7.5 m away from the exhaust pipe and at a height of 1.2 m from the ground.

RESULTS (Cont'n)

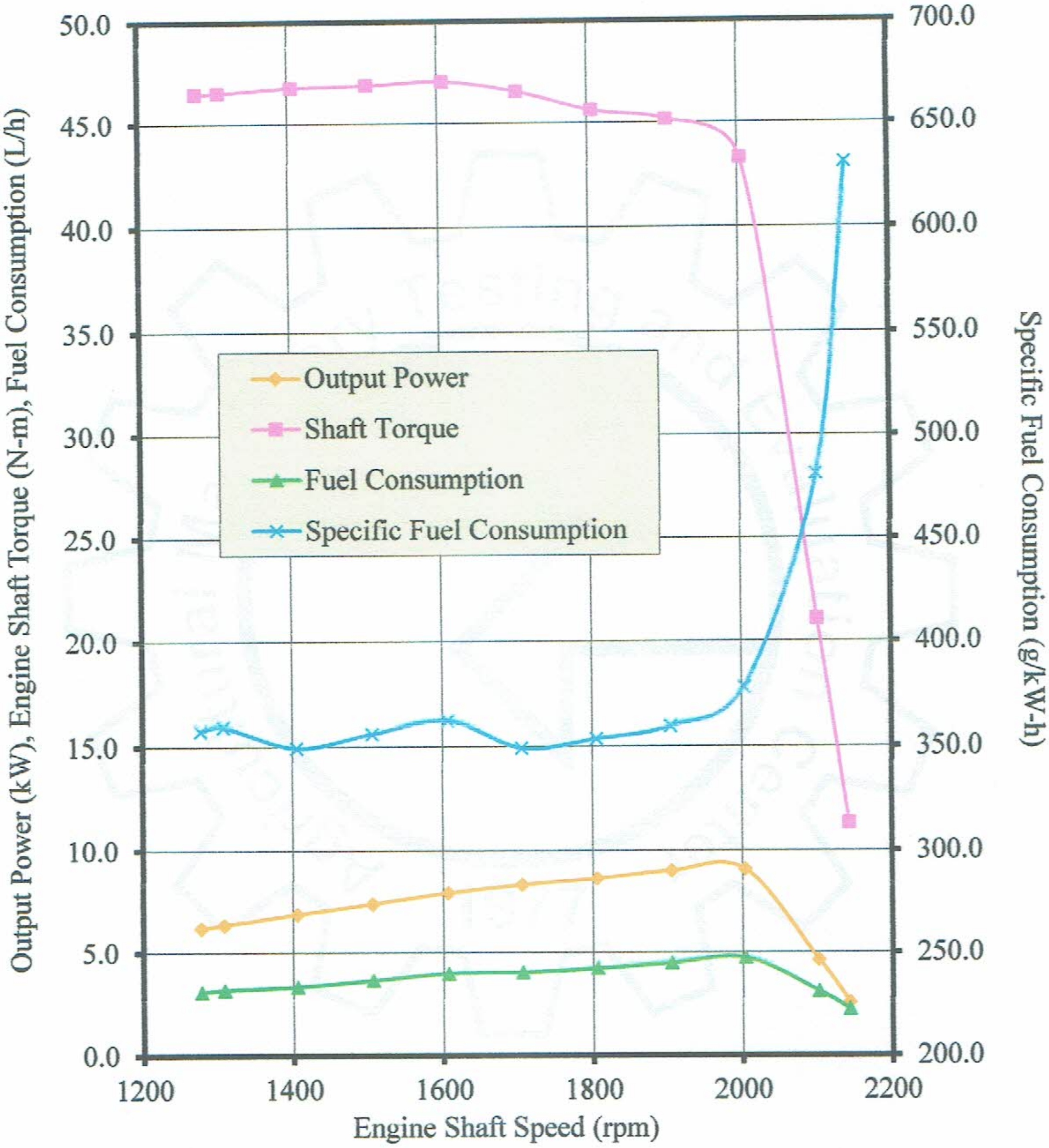


Figure 3. Performance characteristic curves during varying load test of the KRESS KE215R Gasoline Engine.

RESULTS (Cont'n)

Continuous Running Test

Table 4. Test conditions during the continuous running test of KRESS KE215R Gasoline Engine.

Date of test	January 17, 2023
Duration of test, h	5.0
Ambient temperature, °C	32.7
Relative humidity, %	68.1

Table 5. Results of continuous running test of the KRESS KE215R Gasoline Engine set at 77.10% of the rated maximum power.

Time, h	Engine Shaft Speed, rpm	Shaft Torque, N-m	Output Power, kW	Fuel Consumption, L/h	Specific fuel Consumption, g/kw-h	Noise Level, dB(A)	Temperature, °C	
							Engine Oil	Exhaust Air
0.0	1806	47.74	9.02	4.01	322.09	88.7	71.3	551.6
0.5	1809	47.16	8.93	4.13	335.20	89.9	134.6	457.9
1.0	186	46.12	8.72	4.00	332.67	89.7	134.5	405.3
1.5	1807	46.15	8.73	3.88	321.90	88.6	134.2	523.9
2.0	1800	45.97	8.66	4.01	335.97	88.9	133.4	330.3
2.5	1798	45.77	8.61	3.97	334.47	88.3	132.6	361.3
3.0	1802	45.04	8.49	4.15	354.00	89.1	131.1	367.3
3.5	1801	44.90	8.46	4.22	361.53	90.4	131.5	366.3
4.0	1806	45.24	8.55	4.36	369.94	90.2	131.9	360.3
4.5	1802	44.65	8.42	4.32	371.67	89.8	131.5	301.3
5.0	1801	44.04	8.30	4.11	359.35	90.3	131.9	303.3
Ave	**1803**	**45.71**	**8.63**	**4.11**	**345.35**	**89.4**	**127.1**	**393.5**

Note: Engine was set at maximum power it can attain at rated speed. No breakdown occurred. Noise level measured at 7.5 m away from the exhaust pipe and at a height of 1.2 m from the ground.

RESULTS (Cont'n)

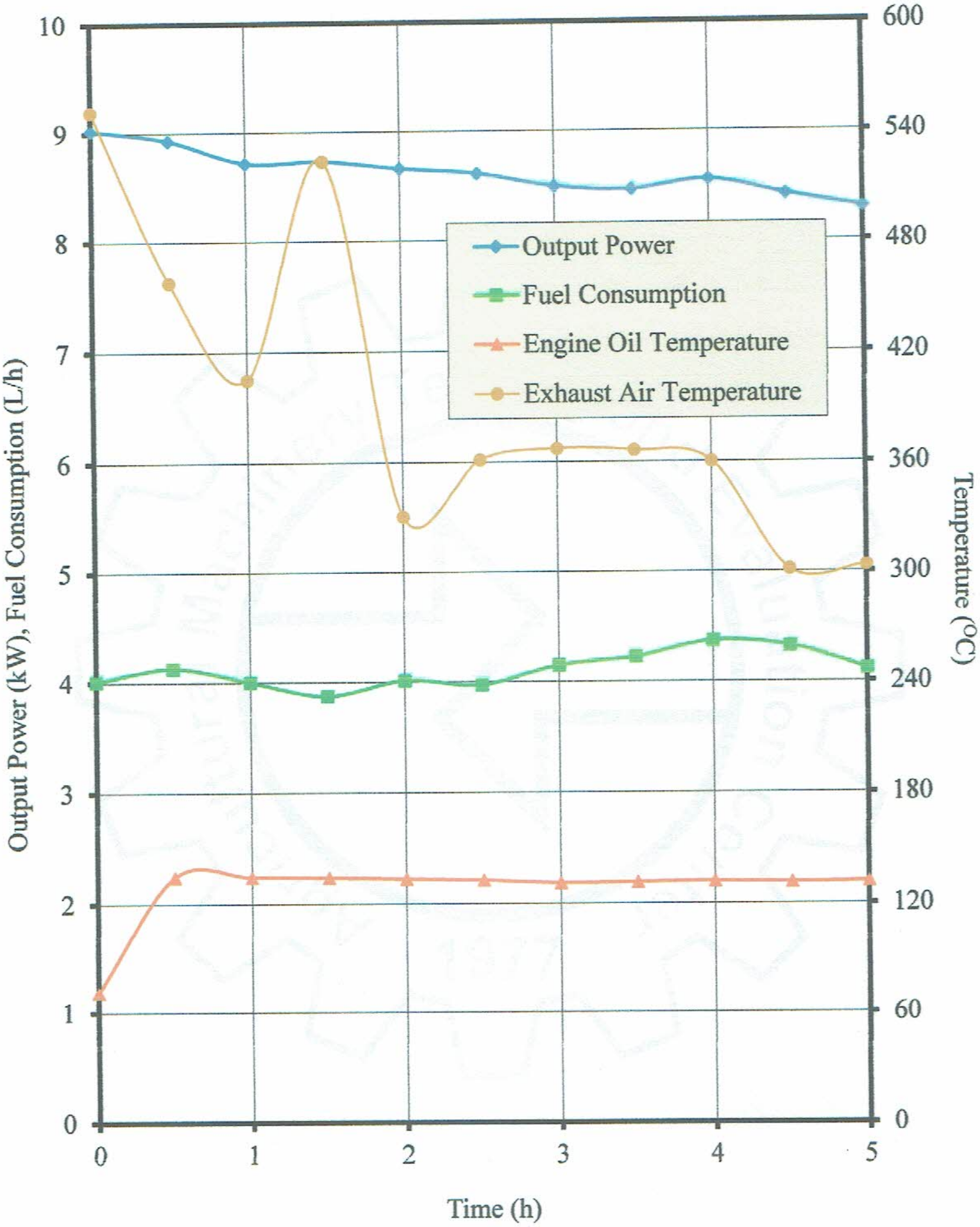


Figure 4. Performance characteristic curves during the continuous running test of the KRESS KE215R Gasoline Engine.

OBSERVATIONS

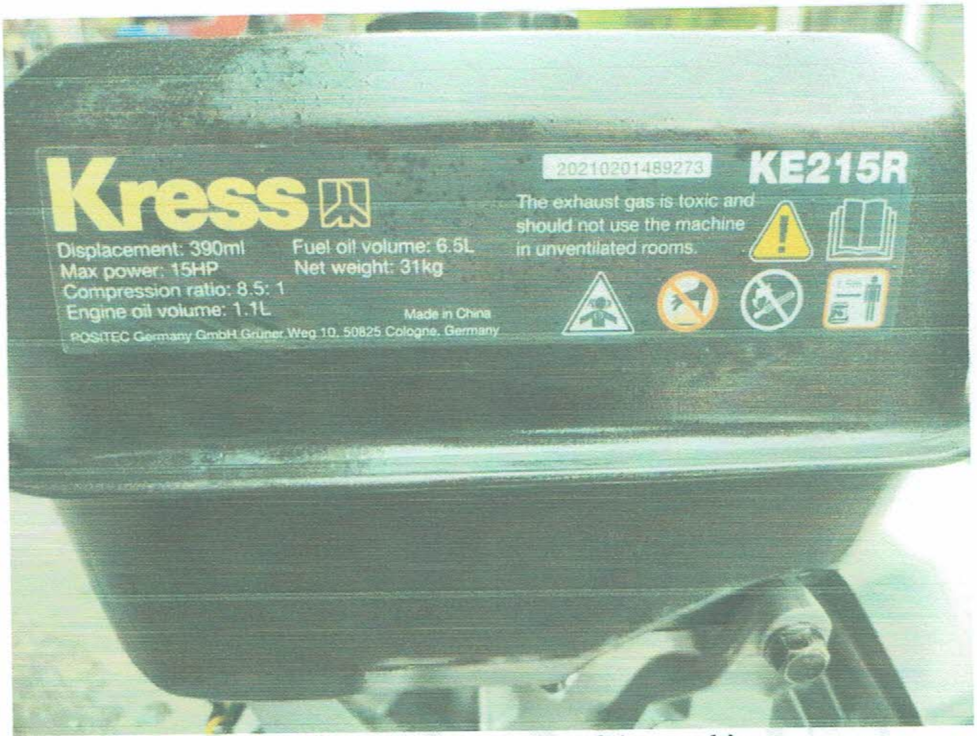


Figure 5. Sticker nameplate of the machine.

Figure 6. Markings, labeling and warning signs of the machine.

OBSERVATIONS (Cont'n)

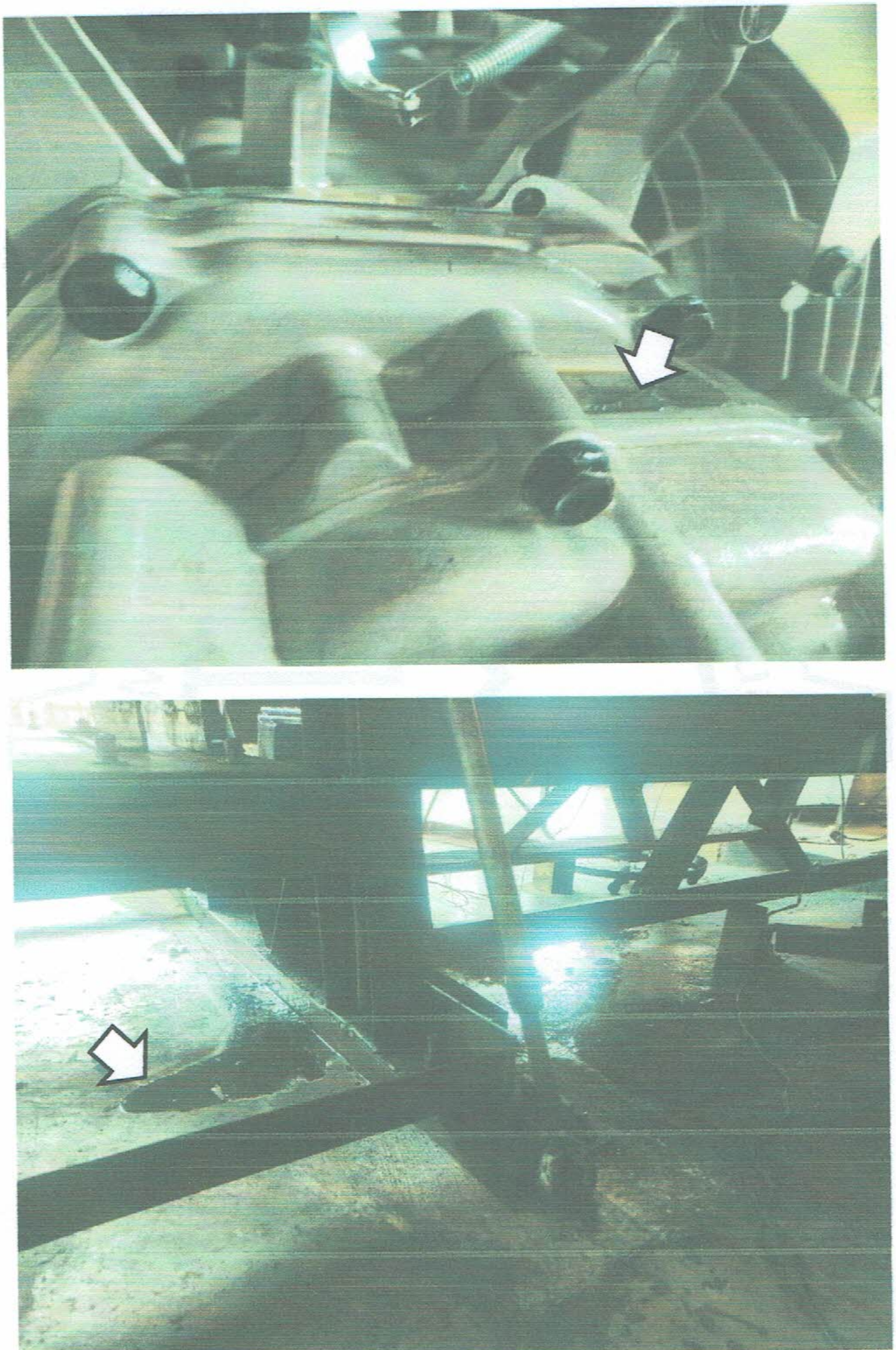


Figure 7. Protruding crank gasket (top) were visible and caused an engine oil leak (bottom) during the 4th hour of continuous running test.

Tested by:

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Test Engineer

Verified by:

KAREN CYBILL F. CABANGLAN
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO
AMTEC Director

JAN 23 2023
Date

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